Global Firms, Local Students
Multinational Presence Shapes College Major Choice

Job Market Paper

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Abstract

Multinational firms shape local economies by changing the returns of skill acquisition and through changes in labor demand. I provide evidence of an alternative channel through which multinational corporations (MNCs) shape human capital formation: by influencing field of study choices prior to labor market entry. Using administrative data from a large public university in Costa Rica, I estimate how local MNC presence affects prospective students' field of study choice. To capture local exposure to MNCs, I develop a measure of multinational firm presence by industry and location, incorporating spatial proximity between firms and university applicants. I estimate a multinomial logit model where students choose a major within grouped field of study categories, allowing for heterogeneous effects to MNC presence across industry and academic discipline pairings. Results show that MNCs impact students decisions, with effects varying by industry and field of study. These findings highlight an important and previously understudied channel through which FDI can shape human capital formation in receiving nations.

1 Introduction

Over the past several decades, foreign direct investments (FDI) have become a driving force in shaping developing economies (Koepke, 2019)[1]. In FDI-receiving countries, multinational corporations (MNCs) not only influence wages, productivity, and technological adoption[2], but also alter the composition of local labor markets (Dustmann and Glitz, 2015; Jude and Silaghi, 2016; Hale and Xu, 2016). A key, but under-explored question is how FDI may influence the skills that individuals choose to acquire through tertiary education (Blomström and Kokko, 2003; Meyer, 2004; Paul and Feliciano-Cestero, 2021). The introduction of MNCs into local economies shift employment opportunities toward two possible ends, low-skill or high-skill jobs. Labor supply, now driven by MNC demand, shifts toward new equilibriums (Blanchard and Willmann, 2016). Employment trends have crowded-out middle-skill opportunities in industrialized economies (Goos and Manning, 2007; Goos et al., 2014) and we should expect similar shifts to follow in developing nations as MNC offer low-skill jobs (Feenstra and Hanson, 1997; Atkin, 2016) and high skill jobs (Javorcik, 2015; Alfaro-Ureña et al., 2021; Setzler and Tintelnot, 2021).

This paper shows that MNCs influence labor markets by shifting students' demand for fields of study, which in turn reshapes the composition of the future high-skill labor force. I study this in Costa Rica, which has experienced significant growth in FDI inflows since 2009. This setting is particularly well-suited because of the many incentives that the country has provided to attract FDI and the detailed administrative level education data from a large public university. Further, Alfaro-Ureña et al. (2021) have provided evidence of an MNC wage premium of up to 9% and a higher demand for highly educated individuals than domestic private firms in Costa Rica. By linking administrative education data with spatial variation in local MNC exposure over time, I examine how multinational presence shapes

[1]FDI reached $916 billion (more than 70% of global flows, in 2022. $208 billion alone entered Latin America and the Caribbean) United Nations Conference on Trade and Development (2023)

[2]See Lipsey (2004); Jung and Mercenier (2009); Dustmann and Glitz (2015) for work on different channels and absorption of FDI

field of study choices among prospective high-skill individuals.

Conceptually, prospective university students view a field of study as targeted skill acquisition which assists them in efforts to be employed in a given industry. For example, to obtain employment in the semiconductor manufacturing industry, they are generally aware that some form of engineering discipline would endow them with the necessary skills. Additionally, students are exposed to local labor market opportunities, which inherently vary by location. As FDI projects increase in quantity and given the greater attractiveness of MNCs relative to domestic private firms, existing foreign firms become more relevant to the student's decision-making process. MNCs may offer additional factors that make them attractive as they may allow for non-pecuniary returns, better future employment opportunities, or possible migration opportunities. As labor supply adjusts to a changing labor market composition, students adjust their skill acquisition decisions accordingly. This adjustment should be considered similarly to adjustments observed after trade shocks (Artuç et al., 2010).

To conduct an empirical analysis, I need data on student's demand for majors (before enrollment equilibrium) and MNC industrial composition and location to identify local labor markets. For data on student demand for major, I use administrative data on university applications to a large public university from 2010 to 2019. For data on MNC firms, I use firms that operate within the MNC-exclusive Free Trade Zone regime in Costa Rica. I then estimate MNC effects on field of study choice using a Multinomial Logit model.

Using an empirical model of field of study choice where students choose a major subject to MNC presence allows me to measure both magnitude and direction of these effects. Students are differentially exposed to MNCs based off of their district of residence and year of application. MNC exposure is captured by a tenure-weighted presence index which is derived from a Gravity model approach using both the firm's and applicant's residence. I then create field of study to industry attachment probabilities matrix so that major choices

have a proportionate and labor-market-relevant response to each industry. This also allows for heterogenous responses across field of study by industry.

Identification in this paper relies on spatial and temporal variation in exposure to multinational firms across districts and years. By including canton (higher regional level) and year fixed effects, the model controls for time-invariant local and national shocks, isolating plausibly exogenous changes in MNC presence. The identifying assumption is that, conditional on individual characteristics and local economic controls, changes in MNC industry composition are unrelated to unobserved determinants of student's education specialization choices. The exposure measure is weighed by field-industry attachment probabilities, so varying changes in relevant industries are also identified from industry exposure across fields within the same region and year.

I find that greater exposure to MNC industries does have an effect on the probability that students choose a given field of study, with effects varying across field of study by industry pairings. Notably, when considering the presence of MNCs that participate in the Manufacturing industry, the largest MNC industry in Costa Rica, a shift from low- to high-MNC presence is associated with an increase in the probability of selecting a major within the STEM and Applied Sciences category by 52.9 percentage points. At the same time, the probability of choosing a major within the Social Science and Professional Studies category decreases by 27.9 p.p. and within the Arts by 25 p.p. Considering another large Costa Rican industry, Administrative Services, the results suggest that a shift from low- to high-MNC presence increases the probability of choosing a major within the Social Sciences and Professional Studies by 39.81 p.p., while seeing a decrease in STEM and Applied Sciences of 8.96 p.p. and in Arts by 30.58 p.p. Results are robust to removal of field of study by industry attachment weights, where industries have equal influence across field of study choices. Conducting a heterogeneity analysis by sex shows no statistically significant differences in the effects of MNC presence across male and female students, which is intuitive given that

foreign direct investment is not targeted along sex lines.

This paper contributes to three main strands of existing literature: (1) interactions between trade and human capital (2) impact of FDI on host economies and (3) determinants of major choice. By identifying and measuring the effects of MNC presence on field of study choices, I help explain a new mechanism through which trade impacts human capital decisions: the presence of multinational firms impacting individual skill specialization decisions. I also add to the growing literature that analyzes the effects of FDI-receiving nations. I demonstrate that FDI's impact on educational attainment extends beyond simple participation, shaping skill acquisition at the intensive margin as reflected in students' decisions regarding their field of study.

The paper is structured as follows. Section 2 describes the relevant literature and highlights contributions. Section 3 describes the institutional details and context of the project in greater detail. Section 4 presents the data. Section 5 describes the empirical approach. Section 6 presents the empirical results and Section 7 concludes.

2 Literature Review

The relationship between international trade and human capital formation has been extensively studied in the endogenous growth literature. Foundational models, such as Romer (1990) and Caballé and Santos (1993), formalize human capital investment decisions as drivers of economic growth. As greater FDI volume enters developing nations, absorptive capacities of receiving nations may dictate efficient productivity gains (Jung, 2020). FDI has complementarities with schooling, where economic growth benefits may depend on existing levels of schooling. Wang and Wong (2011) show that these levels do not need to be necessarily high, with quality being an important determinant over quantity, as was previously shown in Borensztein et al. (1998). From a macroeconomic perspective, FDI has an impact on national average years of schooling attained (Blanchard and Olney, 2017; Kheng

et al., 2017; Dey and Mishra, 2018), and aggregate enrollment counts (Mughal and Vechiu, 2011). In OECD countries, tertiary education has a great influence in attracting inward FDI (Pantelopoulos, 2022). Crucially, Blanchard and Olney (2017) provide evidence that these effects are likely determined by the type of sectoral growth: skill-intensive exports increase schooling.

Looking at empirical evidence, FDI impacts receiving economies through backward linkages between multinationals and domestic firms (Smarzynska Javorcik, 2004), skill upgrading of workers (Jung and Mercenier, 2009), wage premia (Javorcik, 2015; Alfaro-Ureña et al., 2021), technology transfers (Zhuang, 2017), and labor market composition changes (Blanchard and Willmann, 2016). Importantly to the context of this paper, Alvarado et al. (2017) cautions on the efficacy of FDI in driving growth in Latin America, where high-income countries seem to be the only true beneficiaries.

FDI inflows may also contribute to broader structural transformations of local markets, through industry composition and patterns of skill demand in the host economy. Blanchard and Willmann (2016) develop a model of skill acquisition where trade impacts education and employment attainment. They conclude that targeted educational policies would benefit middle-skilled workers when exposed to effects of increased trade. Empirically, Goos and Manning (2007) and Goos et al. (2014) highlight a polarization of jobs toward the skill-distribution extremes; low- and high-skill jobs. There is evidence of negative effects on educational attainment levels stemming from greater trade (Feenstra and Hanson, 1997; Atkin, 2016) and demand for higher-skill labor (Javorcik, 2015; Setzler and Tintelnot, 2021).

This paper also relates to literature of determinants of major choice. A large proportion of existing work has focused on the impact of expected earnings on major choice. Observational studies (Altonji, 1993; Carneiro et al., 2011; Beffy et al., 2012; Hastings et al., 2013; Kirkeboen et al., 2016) and laboratory experiments (Arcidiacono et al., 2012, 2020) suggest that

expected earnings are an important determinant of university major choices.[3] Since students are directly influenced by future expected earnings, the wage premia offered by multinational firms should be a direct influence on field of study choices by students. Availability of information is also an important factor for students' decision-making. Previous research shows that changing expectations of returns to education lead to changes in educational attainment efforts. In work done by Jensen (2010), eighth-grade boys in the Dominican Republic completed 0.20 to 0.35 more years of school, on average, after learning of higher returns to education.

Another important factor influencing individuals' choice of field of study are their endogenous abilities, a phenomenon often referred to as ability sorting. Arcidiacono (2004) confirms that an earning premia exist as a determinant of choice but are not able to fully explain sorting across majors. They state that sorting is mainly due to individual preferences for a given field. Kinsler and Pavan (2015) provide evidence that selection into majors plays a role in generating wage gaps across fields. Intuitively, preferences are subject to their surrounding environment and are subject to change. In this project, individuals are subject to their local labor markets and shifting MNC compositions.

The existing literature contextualizes the choice of major through expected wages and endogenous abilities. This paper contributes to the determinants of major choice by highlighting a different channel: the influence of their surrounding labor market composition. I study how the expansion of multinational firms in developing economies shapes students' field of study decisions through changes in the local labor market industry composition of MNCs.

3 Institutional Details

Costa Rica has a total of five public universities: Universidad de Costa Rica (UCR), Universidad Nacional (UNA), Instituto Tecnológico de Costa Rica (TEC), Universidad Estatal

[3]See Altonji et al. (2012) for a review of both theoretical and empirical existing literature.

a Distancia (UNED), and Universidad Técnica Nacional (UTN). Tuition costs are generally low in Costa Rica, with a university credit costing approximately $30 in 2025.[4] Obtaining a tertiary degree (I only consider bachelor's degrees in this project) will usually take students 5 years on average. Due to data availability, I only observe applications into UNA.

Admissions Process

The admissions process to a public university in Costa Rica is conducted using a decentralized process, where individuals apply directly to each university separately. In order to apply to any public institution, the applicant must take an entry exam meant to evaluate verbal and mathematical reasoning skills. Universities generate an entry score for each prospective student through a weighted average of their performance on the entry exam and their performance over the last two years of their high school education. After receiving their scores, each student is asked to submit their two preferred majors, in order.[5] The institution determines, through budgetary and internal processes, how many seats it will award for each major each year but this is unknown to the student at the time of application.

However, the institutions make public the historic cut-off grades for being admitted into each major. Although this does not guarantee admission simply by having a higher entry score than previous years, it does provide the student with important information as they choose majors. Once the application deadline passes, the university rank-orders each student by major and entry score. The university then admits students into their first choice major going down in order of entry score until there are no more seats available. This constitutes the "cut-off" grade for being admitted into the major, which is essentially the last student to have been accepted. Once this sorting finalizes, students are informed of their application status, and they are allowed to enroll into the institution.

[4] This corresponds to nearly $400 for a full credit load semester. Over 50% of students receive some form of scholarship.

[5] An example from Universidad de Costa Rica can be seen in figure A3.

Free Trade Zone Requirements & Incentives

The Free Trade Zone regime in Costa Rica is a set of incentives and benefits given to firms that realize new FDI projects that meet all requirements. It is mandated by Free Trade Zone Law N7210 and all articles therewithin. Firms that are accepted into the regime establish their activities within determined industrial parks which are strictly designated for firms operating under the FTZ regime. It is possible for firms to obtain authorization from the government to establish operations outside the industrial park but these are rare cases. For simplicity, I remove all firms that are reported as operating outside of the industrial parks in the data.[6]

There is a required minimum investment amount of $150,000 in fixed assets. This investment requirement is less for firms that locate themselves outside the Greater Metropolitan Area (GAM, from the Spanish acronym), being required to a minimum investment amount of $100,000. These investments must be completed in the initial 3 years after the firm has been notified of their acceptance into the regime. Beyond the initial financial investment requirement, firms must also abide by both local and international strict environmental policies and must be able to prove they abide by them.

Firms are allowed to carry out all regular business activities, plus other specific ones outlined by the law, without limiting their overall scope. Firms that provide services can serve both firms in the FTZ and foreign clients. Manufacturing and processing firms can perform a broad range of activities on goods for export/re-export, except for activities prohibited by provisions in the law. Outsourcing of services is allowed without limitations if these services are provided to foreign firms or to firms also operating within the FTZ regime. If services are provided to firms outside the FTZs, only a maximum of 50% of total sales may come from these clients. Similarly, local sale of products by processing firms may represent up to 25% of the firm's total sales, except in the case that the firm is labeled as a "Type f processing

[6]Firms may request special permission to operate outside of the Industrial Park but this is subject to approval by the governing body.

firm", as indicated in subsection (f) of Article 17 of the FTZ Law. Subsection (f) firms may sell 100% of their output in the local market, meaning that they are not required to export in order to benefit from the FTZ regime.

4 Data

MNCs Operating Within FTZ Regime

Firms under this regime are by definition Multinational Corporations. I begin with the entire universe of firms operating within the FTZ regime as of 2023. The data contains each firm's name, location at the province-canton-district[7] level, whether they operate inside the industrial park or not, industries they engage in, and a unique corporate identity number. A crucial part of my identification strategy is the time of entry of each firm, which is missing in this data. To remedy this, I use each firm's unique corporate identity number to identify their earliest date of registry with the Costa Rican government using the Costa Rica National Registry.[8] From this data I select the earliest official document's date possible and assign it to the respective firm.

The economic activity codes are equivalent to the 4-digit International Standard Industrial Classification (ISIC Rev.4). For simplicity, I use the first reported code in the data. I then concord these codes to their 2-digit categories in order to properly map it with the industry by field of study mapping I detail below. This produces 31 unique 2-digit ISIC Rev. 4 categories, with the largest proportion of firms being in Manufacturing (29.5%). The full list can be found in table A3 of the Appendix.

[7] Costa Rica is administratively organized into 7 Provinces, 81 Cantons, and 488 Districts.
[8] Access to the registry can be completed online at http://www.registronacional.go.cr

Education

Data on university applications from the Universidad Nacional (UNA) are sourced directly from the university. Each observation is an anonymized individual application containing the year they applied, their province-canton-district of residence, age, sex, type of high school attended, entry score grade, their first and second choices for major and the respective International Standard Classification of Education (ISCED) category as determined by UNESCO. For UNA, there are 77 unique majors which are categorized into 10 Broad Categories. To make the possible choice set for each student more tractable, I aggregate the possible majors using the ISCED broad categories into three fields-of-study. These are Arts, Writing and Tourism, STEM and Applied Sciences, and Social Sciences and Professional Studies. These are the possible choices each individual will face when completing their application. Table A1 in the Appendix shows these with examples of majors within each category.

The data from UNA expands from 2010 to 2019 and contains 230,162 applications. The sample leans more female (57%), an average age of 19.49 years, and over 75% of applicants come from public high schools. The majority field of study chosen are disciplines on Social Science and Professional Studies (50.7%) and STEM being the second largest (36%). Application by regions logically follow population patterns, where most come from the largest provinces. Greater details are shown in appendix table A4.

In Section 5, I detail how I map industries to the fields of study in my analysis. They are probability attachment weights which capture how likely a given field of study is to have employment in a given industry. To do so, I use the American Community Survey (ACS) for each year of the sample period.[9] The ACS surveys individuals on their education attainment and which industry they are employed in, if any, amongst many other questions. One question asks respondents which field they have a degree in, which using the same ISCED categorization structure as in the education data above, I sort into the three aggregated fields

[9] I am unaware of similar data existing for Costa Rica.

of study.

5 Methodology

Model

In order to estimate how MNC presence impacts field of study choice, I model individuals utility as:

$$U_{idmt} = \beta_m \Gamma_{dmjt} + \gamma_i X_i' + \alpha_c + \alpha_t + \varepsilon_{idmt}, \tag{1}$$

where individual i from district d chooses the field of study m in year t that maximizes their utility. β_m captures the effect of MNC presence on field of study choice. The MNC industry presence index (Γ_{djt}) measures the level of influence that any given industry has on the student. The index itself is weighed by the probability attachment weights. In this way, fields with a higher likelihood of being employed in a given industry receive a more appropriate amount of influence, and vice-versa. Each individual also has demographic characteristics X_i', like their age, sex, entry score, high school type and the local unemployment rate. They are also subject to canton and year fixed effects that absorb unobserved factors that will influence all individuals in the same canton or year. While the fixed effects shift the overall level of utility, they do not affect choice probabilities directly in the multinomial logit regression as the choice depends on differences on utility across alternatives. Individuals are also subject to an idiosyncratic shock ε_{idmt}. The possible field of study choices are those noted in table A2.

The individual choice is estimated using a Multinomial Logit (MNL) model where individuals choose the field of study which gives them the highest utility:

$$P(Y_i = m) = \frac{\overbrace{exp(\beta_m \Gamma_{dmjt} + \gamma_i X_i' + \alpha_c + \alpha_t + \varepsilon_{idmt})}^{\text{Utility of chosen field of study}}}{\underbrace{\sum_{m'} exp(\beta_{m'} \Gamma_{dm'jt} + \gamma_i X_i' + \alpha_c + \alpha_t + \varepsilon_{idm't})}_{\text{Utilities across all possible field of study choices}}} \tag{2}$$

This gives the probabilities of choosing field of study m as determined by the utility function above. The coefficients of interest are the β_m, which capture the effects of MNEs through their industry to choose a field of study m. Specifically, Γ_{dmjt} is an index of the presence of an industry j in year t weighed by the distance between firms and individual i. This ratio is then multiplied by the attachment probabilities between field of study m to industry j for each year t. These coefficients quantify the extent to which students' field of study choices respond to influences of MNCs in the local labor market.

MNC Presence Index

In order to create an index that captures how individuals experience the presence of MNCs, I use the following gravity model structure:

$$\Gamma_{dmjt} = p_{mjt} \times \left(\sum_{d'} \sum_{j} \frac{\text{Tenure}_{k(j),d't}}{exp(\text{dist}_{dd'})} \right) \tag{3}$$

where the numerator $\text{Tenure}_{k(j),d't}$ is the tenure of firm k in industry j in district d' operating in year t. A firm's tenure is used as a measure of the intensity of its local presence, under the assumption that longer-established firms are more salient to individuals because they have been present for more years. Each year applicants are exposed to a different stock of firms, and thus different industry compositions, as new firms are likely to enter in a given year. The denominator $exp(\text{dist}_{dd'})$ serves as the distance weight, where it measures the distance (in kilometers) from district d (residence of applicant) to district d' (residence of firm), using each district's centroid. Distance is exponentiated to account for circumstances

where the applicant and firms share the same district and distance is measured as zero[10] and to penalize larger distances between firms and students.

Given how the MNC presence index is constructed, an increase of one unit in the index can potentially arise from two events: (1) individuals relocate closer to the firms or (2) new firms enter the economy. Because the data are repeated cross-sections, there is no possibility of migration. Without migration as a mechanism, this leaves new firms entering or aging as the only possibility of an increase. Given that the index is also weighed by spatial distance, one new firm does not necessarily equal one unit increase in the index. An increase of one unit should be interpreted as an additional firm-year in the student's district of residence. For increased clarity, the results are interpreted using changes in intensity of the index. These are shifts from low- to high-presence of industries for the individual.

Average Marginal Effects

To better interpret the effects of MNCs on prospective students' field of study choices, I compute the average marginal effects (AMEs) from the above model. While the multinomial logit model coefficients β_m describe how industry-specific presence Γ_{dmjt} affects the latent utility of choosing a specific field of study m, they are not directly interpretable in terms of choice probabilities due to the non-linear form of the logit model.

AMEs allow for a direct interpretation by transforming the estimated logit utilities into changes in predicted probabilities. Specifically, the marginal effect of an industry presence variable Γ_{djt} on the probability that individual i chooses field of study m is given by:

$$\frac{\partial P_{im}}{\partial \Gamma_{djt}} = P_{im}\left(\beta_m - \sum_{m'} P_{im'}\beta_{m'}\right), \tag{4}$$

where P_{im} is the predicted probability of individual i chooses field of study m, and the

[10] $exp(0) = 1$, which then gives full weight to immediately located firm-industries

summation term reflects the average effect of Γ_{dmjt} across all possible field of study choices. This captures the reallocation of probability mass across competing alternatives from a change in local MNC-industry composition.

I compute the average marginal effects, which essentially computes this for each individual and averages over the sample:

$$\text{AME}_{mj} = \frac{1}{N} \sum_{i=1}^{N} \frac{\partial P_{im}}{\partial \Gamma_{djt}}. \tag{5}$$

These AMEs represent the expected change in the probability of choosing field of study m with a one-unit increase in the presence index of industry j, holding all else equal. Because these are direct predictions on changes in probability, there is no reference category and all probabilities should sum to zero. The coefficients shown in table A6 are in percentage points.

Education to Industry Mapping

A crucial detail for this project is the ability to map individual industry effects to possible fields-of-study. Given that any university degree can potentially find employment across multiple industries, it is necessary to address how I can differentiate across education field by industry pairings. Using the American Community Survey, I leverage two crucial questions asked. Each individual is asked if they have a degree and if so, in which field. Additionally, they are asked if they are employed and if they answer yes, then they are assigned an industry. I concord the degree/field question using the same ISCED categorization as is used for the university application data. Reported industries of employment use the North American Industry Classification System (NAICS) with varying amount of detail by individual. I concord them to their 2-digit classifications and then concord those to their ISIC Rev.4 equivalents so they match up with the MNC firm data. This creates a harmonized dataset, which allows me to create a probabilistic attachment weight from each field of study to observed industries.

These attachment weights allow me to assign probabilistic relevance of each field of study to each existing industry. This reflects the fact that, although any given industry, for example, may hire an individual with a degree in chemical engineering they may also find employment with a financial firm although that is less likely. It creates differential effects across the education by industry pairings choice set. The mapping can be found in table A5 and the within-sample distribution can be found in figure A5.

6 Estimation Results

I begin by examining whether multinational presence affects the extensive margin of higher education participation and find no evidence of changes in application behavior. I then turn to the intensive margin, estimating the effects of local MNC presence on students' field of study choices. To aid interpretation, I illustrate how these estimates translate into reallocation across fields as exposure increases from low to high levels. Finally, I show that these effects attenuate sharply with distance, highlighting the highly localized nature of the exposure mechanism.

6.1 Extensive Margin Effects on Application Behavior

A natural first question is whether multinational presence affects students' decisions to apply to university at all. I test this by estimating the effect of local MNC exposure on application behavior. To do this, I regress the district count of applications on the MNC industry presence index and aggregated district-level control variables.

Overall, there seems to be no evidence of extensive margin effects of MNC presence on applications. That is to say, multinational firm entry and presence do not seem to push more individuals to apply into tertiary education. The absence of extensive margin effects strengthens the interpretation of the intensive margin results below, as it implies that multinational presence reallocates students across fields rather than altering the composition of

the applicant pool.

I test this result using four different specifications: (1) OLS with district and year fixed effects, (2) OLS with canton and year fixed effects, (3) Poisson Pseudo Maximum Likelihood model with canton and year fixed effects (4) OLS with log count of applications with district and year fixed effects. The estimated effects are similar in sign and magnitude across all four specifications. Results are presented in Appendix table HERE and in Appendix table HERE.

The absence of extensive-margin effects helps isolate the mechanism at the field-choice margin. In the absence of changes in application behavior, shifts in field of study choices can be interpreted as reallocations within a stable applicant pool, reducing concerns that results are driven by selection into higher education.

6.2 Average Marginal Effects of Multinational Presence on Field of Study Choice

I now focus on how local multinational presence affects the composition of applicants across fields of study. These results look at intensive margin effects of multinational industry presence on field of study choice. Results in table A6 present the Average Marginal Effects (AMEs) of MNC presence, measured at the district-level for all industries. Each column shows how an increase in the presence of a given industry shifts the likelihood of applicants choosing a field of study, conditional on the set of controls mentioned above. These results should be interpreted as the effect of an additional firm-year in the student's district of residence. This could be from the entry of a new firm or an existing firm maturing by one year.

I focus on Manufacturing and Administrative Services industries because they are among the most economically significant industries in the country, while also representing two distinct segments of the economy: production and services. Together, these sectors account for

a substantial share of multinational activity and labor demand, making them central to understanding how MNC presence shapes educational choices. Moreover, as the country seeks to transition toward higher-return sectors such as services, comparing the effects of both provides a natural lens through which to assess how multinational exposure may differentially influence human capital formation.

Manufacturing is a significant industry in Costa Rica as it relates to FDI inflows. Over the sample period (2009 to 2019), it represents the majority share of all FDI inflows (36.8%), and ranges from 0.67% to 3.05% of GDP. Being the largest industry (as measured in inflow dollars), observed effects of this industry are of economic interest.

An increase in the presence of manufacturing MNCs leads to a significant rise in the probability of choosing a major in STEM and Applied Sciences, which are accompanied by declines in the Arts and Social Science programs. To be more precise, an increase of one unit in the manufacturing MNC presence index increases the probability of choosing a discipline in STEM & Applied Sciences by 84 percentage points. This is accompanied by a decrease in the probability of choosing a discipline in the Arts and in Social Sciences by 55 and 29 percentage points, respectively. This pattern may indicate that the presence of production and technology intensive firms raises the perceived returns from more technical human capital fields, steering students toward science and engineering disciplines.

By contrast, an increase in the presence of MNCs in the Administrative Services industry is associated with a shift away from STEM and Applied Science (a probability decrease of 100.6 percentage points) toward Social Sciences and Professional Studies (a probability increase of 74.3 percentage points) and Arts (increase of 32.12 percentage points). This suggests that firms more likely to engage in managerial or organizational services encourage human capital investment in business-related or professional skills over technical ones.

Some estimated average marginal effects exceed probability bounds, reflecting strong directional responses rather than literal changes in choice probabilities. For this reason, the

magnitude of these effects is more naturally interpreted through the within-sample shifts from low to high exposure presented in the following subsection.

6.3 Reallocation Across Fields from Low to High MNC Presence

While average marginal effects summarize direction and magnitude, they can be difficult to interpret in terms of actual changes in students' choices. Further, using shifts from low- to high-MNC presence exposure helps to better interpret within-sample shifts in field of study choice. These are within probability bounds and maintain the identified shifts in field of study choices. These results are presented in Appendix table A7 and shown graphically in Appendix figure A7. To be consistent, the primary analysis will be constrained to the Manufacturing and Administrative Services industry. The analysis is easily extended to the other reported industries.

Looking at the Manufacturing industry shown in figure 1 below, the results suggest that a shift from low- to high-Manufacturing industry exposure is associated with an increase of 52.9 percentage points in the probability of choosing a major within the STEM and Applied Sciences field of study category. At the same time, there is an associated decrease in the probability of choosing a major within the Arts category (25.1 p.p.) and the Social Sciences and Professional Studies (27.9). This pattern is consistent with a mechanism in which students adjust their educational choices in response to perceived labor market opportunities associated with manufacturing activity, favoring technically oriented fields at the expense of more general disciplines.

Figure 2 below shows results for the Administrative Services industry. A shift from low- to high-Administrative Services industry exposure is associated with an increase of 39.8 percentage points in the probability of choosing a major within the Social Sciences and Professional Studies category. The Arts and STEM categories see a proportional decrease in the likelihood of choosing a major of 30.8 p.p. and 9 p.p., respectively.

Taken together, these results point to a strong demand-side response by students toward fields more closely aligned with the respective industries. The magnitude of these shifts highlights manufacturing multinationals as a powerful force shaping the composition of the future skilled workforce toward technically oriented fields. At the same time, administrative services multinationals incentivize future skilled workers to acquire business oriented skills. Any structural change is conditional on supply by universities being able to properly adapt as well.

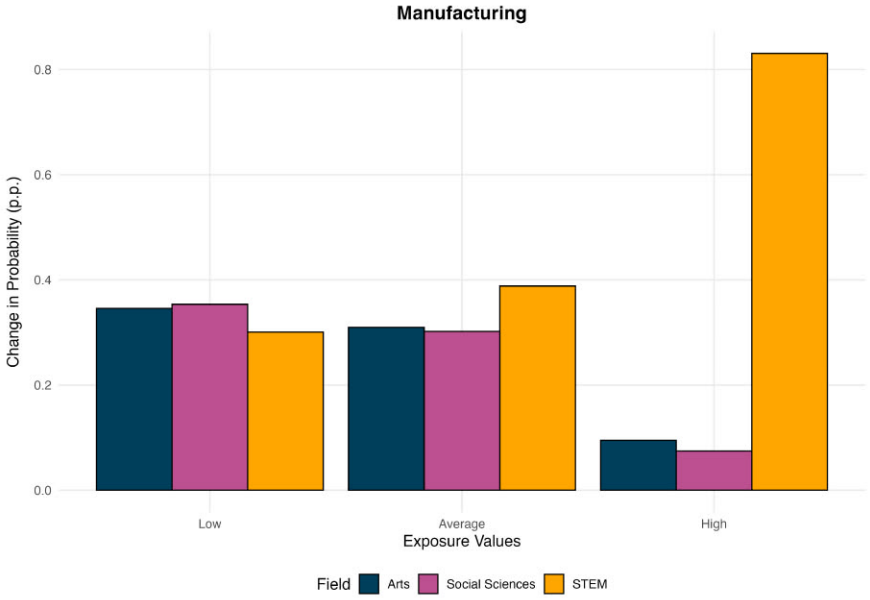


Figure 1: Manufacturing Industry Effects from Low- to High-Exposure Shift

6.4 Spatial Decay and the Local Nature of Exposure Effects

If multinational presence influences educational choices through local exposure, its effects should attenuate with geographic distance from firm locations. To test this implication, I estimate the effects of multinational presence across discrete distance bins between firms and applicants.

Figure A8 shows a pronounced spatial decay in these effects. The estimated impacts on field-of-study choice are concentrated within approximately 5 kilometers of firm locations

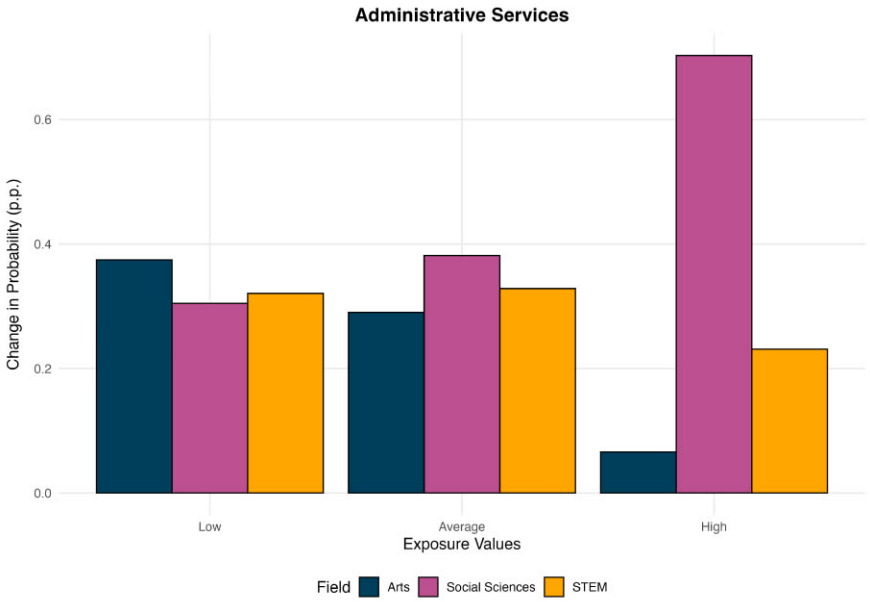


Figure 2: Administrative Services Industry Effects from Low- to High-Exposure Shift

and decline sharply with distance, becoming statistically indistinguishable from zero beyond this range. This hyper-local pattern is difficult to reconcile with broad regional shocks and instead points to a mechanism rooted in localized exposure, such as firm visibility, information transmission, or geographically bounded networks.

Robustness Tests

No Attachment Probabilities

To evaluate the role of the attachment probability weights in influencing the estimated effects, I re-estimate the model without weighing the MNC industry index. This removes the direct linking between industry and field of study pairings. Essentially, individuals' choices make no distinction between fit of field of study and industry. The model, without weights, assigns equal exposure between industry and field of study, likely diluting the observed connections between specific industries and disciplines.

Results of this estimation are shown in table A8 and figure A10. The resulting coefficients remain consistent with the baseline estimates, suggesting that the main findings are not

driven by the weighing mechanism. Although the removal of weights yields attenuated coefficient estimates, the direction of the effects are largely preserved, supporting their consistent economic interpretation.

Heterogeneity Tests

Differences by Sex

If exposure to MNC activity has an effect on perceptions of skill demand, these effects may manifest differently for men and women. To explore the possibility of differential effects by sex, I estimate MNC presence marginal effects by this dimension. Existing research highlights the existence of differences between male and female students when making educational attainment decisions. Results can be found in figure A11 in the Appendix.

The results show no clear differences in the effects between males and females. Coefficients are similar across gender, suggesting that the relationship under study is not driven by sex-specific responses. This is consistent with the expectation that FDI does not systematically target men or women differently, making the lack of gender differences logical.

Differences by Cohort

I also examine heterogeneity across cohorts to capture potential changes over time in how individuals respond to MNC presence. Earlier cohorts may have faced limited exposure to multinational activity or may lack an initial understanding of MNC labor demand. At the same time, later cohorts may have adjusted expectations and educational choices in response to experiencing a more pronounced multinational firm presence. Cohort variation can reveal whether the estimated effects reflect a transitory adjustment or a persistent shift in the way students perceive returns to different fields of study. Results for this margin can be found in figure A12 in the Appendix.

Estimates across cohorts appear to be consistent in the Manufacturing industry. Other

industries appear to show a diminishing effect on application decisions, likely indicating adaptation by both the market and students decisions.

Differences by Greater Metropolitan Area Classification

Finally, regional heterogeneity is scrutinized to understand how local context impact the relationship between multinationals and field of study choices. The specific regions under scrutiny are districts within the Greater Metropolitan Area and those outside of it. The greater density of population found in the nation's metropolitan area and agglomeration of MNCs may give further insight on direct effects of firm presence on educational attainment choices. Results can be found in figure A13 in the Appendix.

For individuals residing within the Greater Metropolitan Area, coefficients closely mirror those of the pooled estimates, confirming that the main effects are largely identified within urban areas. Coefficients in non-metropolitan areas are much larger and less precise, which reflects limited variation in MNC exposure rather than meaningful structural differences.

7 Conclusion

Human capital attainment is an important aspect of economic growth and FDI has become a popular choice for developing economies as a growth strategy. Educational attainment decisions are influenced by many factors, such as expected wages or endogenous ability sorting. Local labor market composition may also directly influence educational attainment decisions.

This paper shows that foreign direct investment influences educational attainment at the intensive margin, by shaping the types of skills students acquire through their choice of majors when applying to university. Using detailed administrative data and spatial and temporal variation in local exposure to multinational firms, the results indicate that greater MNC presence leads students to shift toward fields more closely aligned with expected skill

demand. These effects are economically meaningful and robust to multiple specifications.

A central insight of this paper is that the influence of multinational firms on human capital formation is sharply spatially bounded. The distance-decay results show that the effects of multinational presence on students' field of study choices are concentrated within a narrow geographic radius and disappear beyond approximately five kilometers. This hyper-local pattern reinforces the interpretation that multinational activity shapes educational decisions through localized exposure rather than broad labor market shifts, and complements the finding that multinational presence reallocates students across fields without affecting overall participation in higher education. Together, these results highlight the importance of place-based exposure in understanding how global firms shape local skill formation.

Heterogeneity analyses suggest that the overall impact of FDI on education is broadly consistent across groups. There are no systemic differences by sex, which aligns with the notion that FDI does not target male or female labor specifically. Cohort results remain stable over time, pointing to a persistent relationship between multinational activity and educational specialization. Differences across space are driven primarily by metropolitan areas, where most of the identifying variation arises; in contrast, effects outside these areas are highly imprecise due to limited variation in exposure.

Taken together, the evidence suggests that FDI contributes to shaping the future skill composition of the domestic workforce by influencing educational investment decisions. This highlights a channel through which global integration can affect long-run human capital formation, beyond short-term employment effects and complementary structural composition changes. Understanding these dynamics is essential for policymakers seeking to leverage FDI as part of a broader strategy for inclusive and skill-oriented development.

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A Appendix

Table A1: Classification of Broad Fields Using ISCED-2013 with Examples

Education: Pedagogy, Specific Topics in Primary and Secondary Teaching[a]
Arts & Humanities: Music, Literature, History, Dance, Languages
Social Sciences: International Relations, Economics, Psychology, Sociology
Business Administration: Business Administration, International Commerce
Natural Sciences, Math & Statistics: Industrial Chemistry, Biology, Geography
Information Technologies: Information Systems
Engineering: Topography, Forestry Sciences, Industrial Bioprocesses
Agriculture & Veterinary: Agronomy, Veterinary Medicine, Aquaculture
Health: Social Work, Medicine, Counseling
Services: Tourist Recreation, Sustainable Tourism Business Management

[a] Topics include Social Studies, Mathematics, English, etc.

Table A2: Aggregation of ISCED Broad Categories

STEM & Applied Sciences: Natural Sciences, Engineering, Information Tech., Agriculture, Health
Social Sciences & Professional Studies: Business Admin., Social Sciences, Education
Arts, Writing, & Culture: Arts & Humanities, Tourism[a]

[a] Tourism is not the official category, but rather "Services". It is renamed due to the only majors in this category are some related to tourism activities.

Table A3: Firm Quantity & ISIC4 (2 Digit) Codes

2-Digit Code	ISIC4 Description	Count
	Manufacturing (103 firms)	
10	Manufacture of food products	20
11	Manufacture of beverages	2
13	Manufacture of textiles	2
16	Manufacture of wood and of products of wood and cork, except furniture	2
18	Printing and reproduction of recorded media	1
20	Manufacture of chemicals and chemical products	10
21	Manufacture of basic pharmaceutical products and pharmaceutical preparations	2
22	Manufacture of rubber and plastics products	8
23	Manufacture of other non-metallic mineral products	2
24	Manufacture of basic metals	2
25	Manufacture of fabricated metal products, except machinery and equipment	7
26	Manufacture of computer, electronic and optical products	5
27	Manufacture of electrical equipment	2
32	Other manufacturing	33
33	Repair and installation of machinery and equipment	5
	Electricity, Gas, Steam and Air Conditioning (14 firms)	
38	Waste collection, treatment and disposal activities; materials recovery	14
	Wholesale and Retail Trade (13 firms)	
46	Wholesale trade, except of motor vehicles and motorcycles	13
	Transportation and Storage (15 firms)	
52	Warehousing and support activities for transportation	14
56	Food and beverage service activities	1
	Information and Communication (78 firms)	
61	Telecommunications	1
62	Computer programming, consultancy and related activities	64
63	Information service activities	13
	Real Estate Activities (47 firms)	
68	Real estate activities	1
70	Activities of head offices; management consultancy activities	7
71	Architectural and engineering activities; technical testing and analysis	24
72	Scientific research and development	5
73	Advertising and market research	4
74	Other professional, scientific and technical activities	6
	Administrative and Support Service Activities (77 firms)	
78	Employment activities	1
82	Office administrative, office support and other business support activities	76
	Education (1 firm)	
85	Education	1

Table A4: Demographic Summary Statistics

Demographic Information			
	Mean	Std. Dev.	
Age	19.49	(2.65)	
Application Score	568.74	(110.87)	
Female	0.57		
STEM[a]	0.36		
GAM[b]	0.55		
Public High School	0.77		
Geographical Information			
Province	(%) of Applications	No. of Cantons	No. of Districts
San José	30.67	20	108
Alajuela	19.82	16	94
Heredia	16.64	10	41
Cartago	8.45	8	48
Puntarenas	8.24	11	57
Guanacaste	10.02	11	58
Limón	6.17	6	28
Observations		230,162	

[a] First choice was a STEM major.
[b] Applicant lives within the Greater Metropolitan Area.

Table A5: Mapping of Degree Fields to ISCED Broad Categories

degfield	degfield label	ISCED Broad Category	Broad Label Category
11	Agriculture	8	Agriculture, Forestry, Fisheries, and Veterinary
13	Environment and Natural Resources	8	Agriculture, Forestry, Fisheries, and Veterinary
14	Architecture	7	Engineering, Manufacturing, and Construction
15	Area, Ethnic, and Civilization Studies	3	Social Sciences, Journalism, and Information
19	Communications	3	Social Sciences, Journalism, and Information
20	Communication Technologies	6	Information and Communication Technologies
21	Computer and Information Sciences	6	Information and Communication Technologies
22	Cosmetology Services and Culinary Arts	10	Services
23	Education Administration and Teaching	1	Education
24	Engineering	7	Engineering, Manufacturing, and Construction
25	Engineering Technologies	7	Engineering, Manufacturing, and Construction
26	Linguistics and Foreign Languages	2	Arts and Humanities
29	Family and Consumer Sciences	1	Education
32	Law	4	Business, Administration, and Law
33	English Language, Literature, and Composition	2	Arts and Humanities
34	Liberal Arts and Humanities	2	Arts and Humanities
35	Library Science	3	Social Sciences, Journalism, and Information
36	Biology and Life Sciences	5	Natural Sciences, Mathematics, and Statistics
37	Mathematics and Statistics	5	Natural Sciences, Mathematics, and Statistics
38	Military Technologies	7	Engineering, Manufacturing, and Construction
40	Interdisciplinary and Multi-Disciplinary Studies (General)	0	Generic
41	Physical Fitness, Parks, Recreation, and Leisure	10	Services
48	Philosophy and Religious Studies	3	Social Sciences, Journalism, and Information
49	Theology and Religious Vocations	3	Social Sciences, Journalism, and Information
50	Physical Sciences	5	Natural Sciences, Mathematics, and Statistics
51	Nuclear, Industrial Radiology, and Biological Technologies	5	Natural Sciences, Mathematics, and Statistics
52	Psychology	3	Social Sciences, Journalism, and Information
53	Criminal Justice and Fire Protection	3	Social Sciences, Journalism, and Information
54	Public Affairs, Policy, and Social Work	3	Social Sciences, Journalism, and Information
55	Social Sciences	3	Social Sciences, Journalism, and Information
56	Construction Services	7	Engineering, Manufacturing, and Construction
57	Electrical and Mechanic Repairs and Technologies	7	Engineering, Manufacturing, and Construction
58	Precision Production and Industrial Arts	7	Engineering, Manufacturing, and Construction
59	Transportation Sciences and Technologies	7	Engineering, Manufacturing, and Construction
60	Fine Arts	2	Arts and Humanities
61	Medical and Health Sciences and Services	9	Health and Welfare
62	Business	4	Business, Administration, and Law
64	History	2	Arts and Humanities

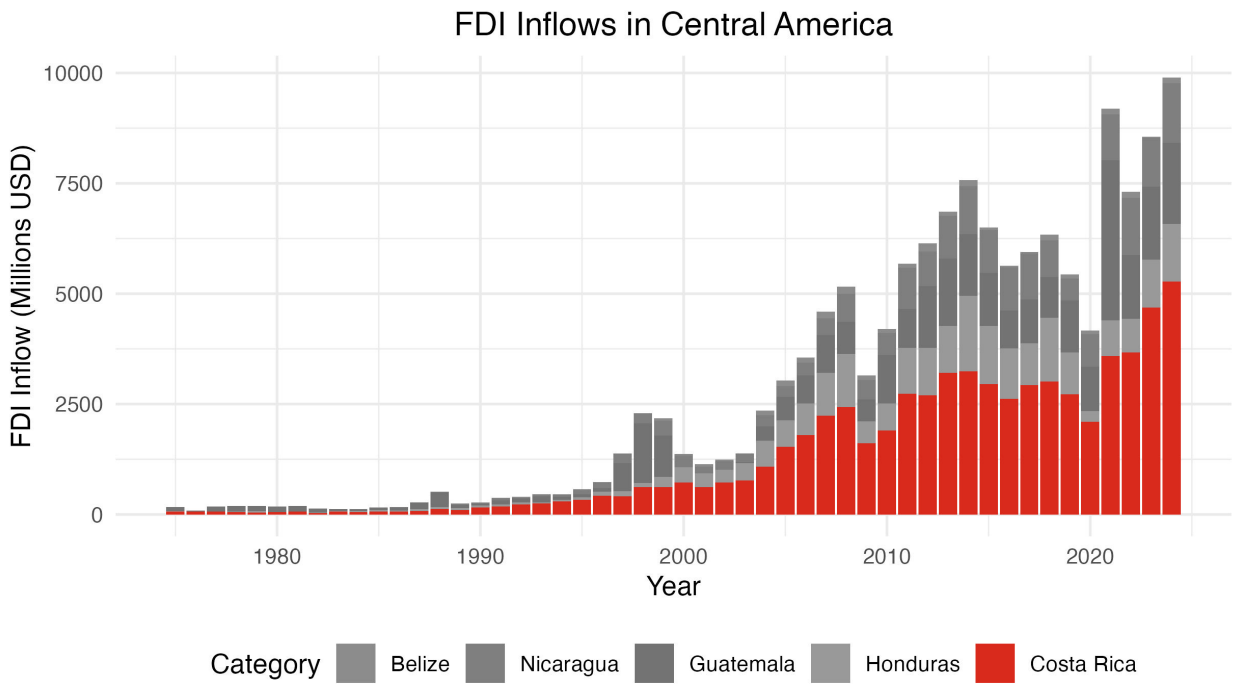


Figure A1: Share of Costa Rica FDI Inflows into LATAM

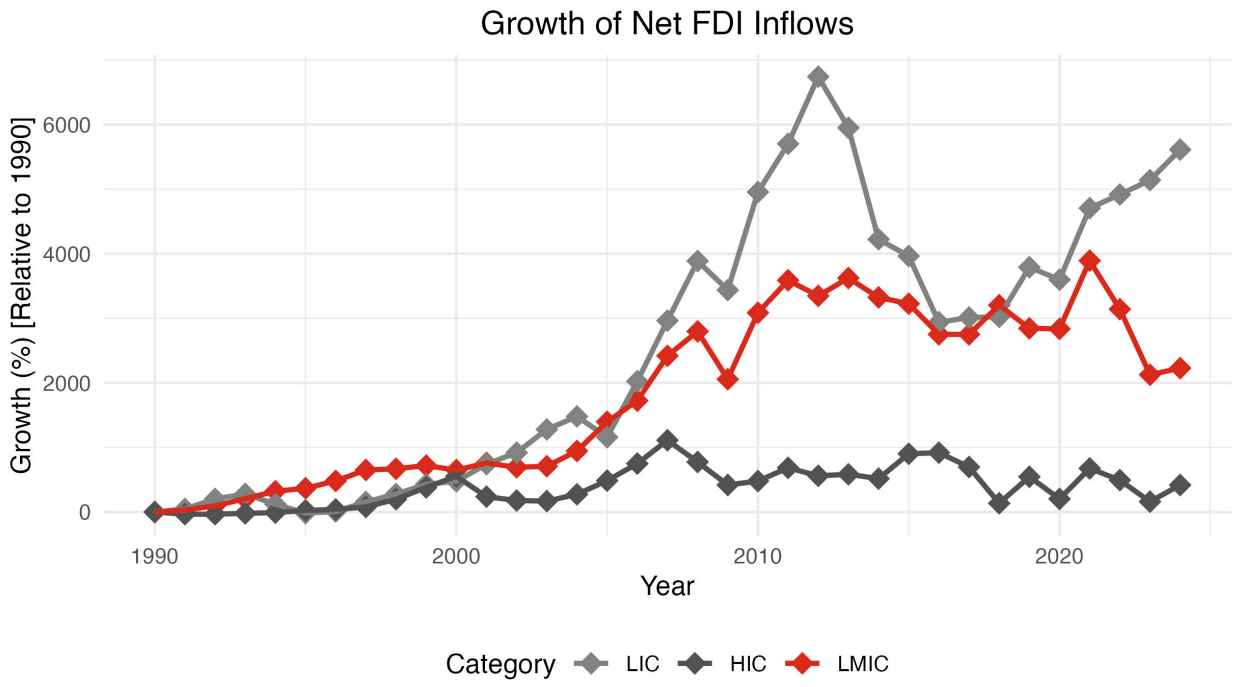


Figure A2: FDI Inflows by LMICs (World Bank)

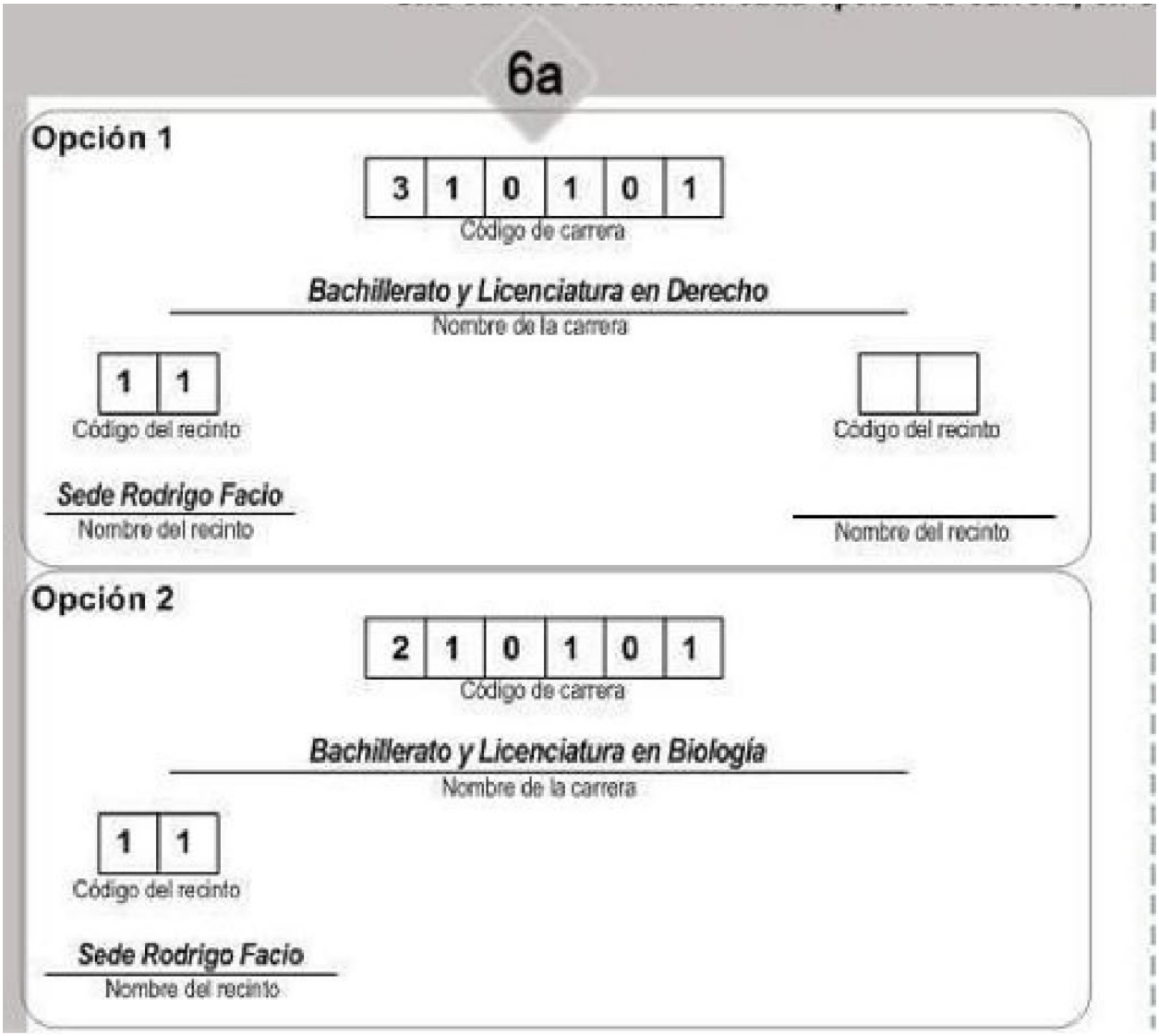


Figure A3: Example of Rank Order of Majors Application

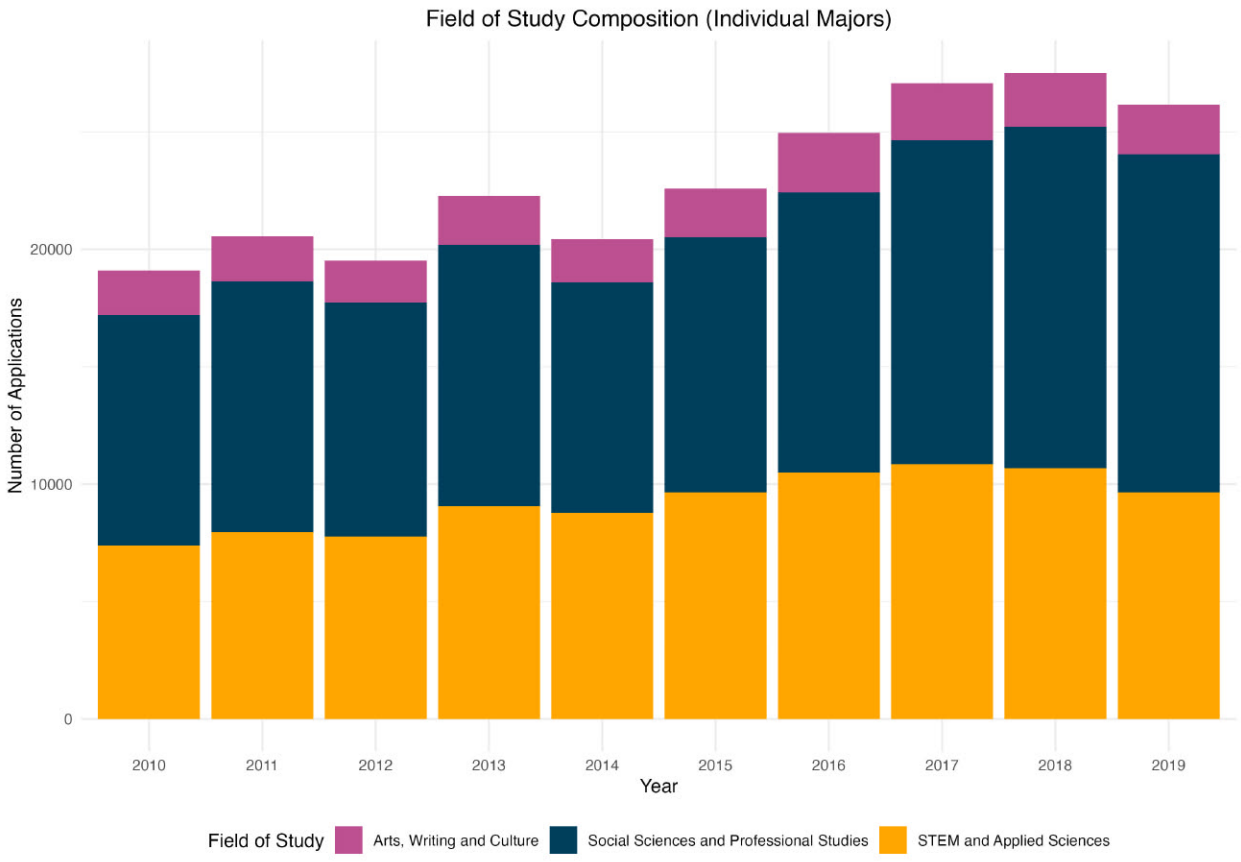


Figure A4: Distribution of Applications by Field of Study

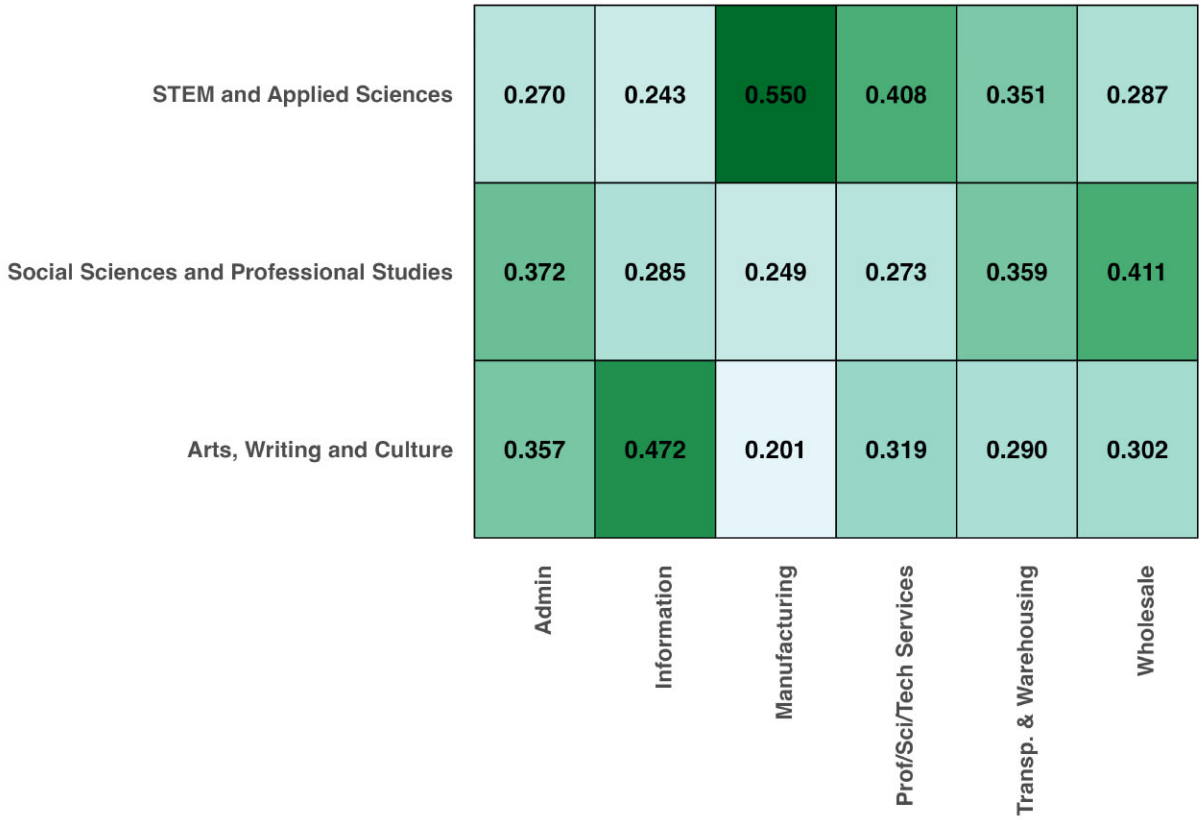


Figure A5: Within Sample Distribution of Probability Attachment Weights Between Field of Study and Industry

Table A6: MNC Industry Presence Effects on Field of Study Choice

	Field of Study Choice		
	Arts, Writing & Culture	STEM & Applied Sciences	Social Science & Prof. Studies
Administration	0.3212	-1.064***	0.743***
	(0.2104)	(0.382)	(0.2096)
Information	1.0349**	-0.6043	-0.4307
	(0.4261)	(0.6701)	(0.4074)
Manufacturing	-0.5566***	0.8408***	-0.2842**
	(0.2038)	(0.3158)	(0.1293)
Prof. Science & Tech. Services	-0.2331	0.1919	0.0412
	(0.1972)	(0.3049)	(0.1614)
Transportation & Warehouse	0.0118	1.4516***	-1.4633**
	(0.3755)	(0.5395)	(0.6348)
Wholesale	3.8000*	-7.7612**	3.9612***
	(1.9937)	(3.3693)	(1.5092)
Controls	✓	✓	✓
Canton FE	✓	✓	✓
Year FE	✓	✓	✓
Applications	20,985	92,273	116,904
Pseudo-R2	0.0436	0.0436	0.0436

Statistical significance is displayed as * $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$
Field of Study Fields grouped up as shown in table A2.
Standard errors are clustered at the District level
By definition of AMEs, each row (industry) coefficients should sum to zero

Table A7: Predicted Field-of-Study Choice by Exposure Level

Industry	Field	Low	Average	High	High − Low
Manufacturing	Arts	0.346 (0.006)	0.310 (0.012)	0.095 (0.054)	-0.251 (0.054)
	STEM	0.301 (0.005)	0.388 (0.029)	0.831 (0.105)	0.530 (0.105)
	Social Sciences	0.354 (0.006)	0.302 (0.019)	0.075 (0.052)	-0.279 (0.052)
Administrative Services	Arts	0.375 (0.012)	0.290 (0.017)	0.066 (0.029)	-0.308 (0.031)
	STEM	0.321 (0.012)	0.328 (0.019)	0.231 (0.131)	-0.090 (0.132)
	Social Sciences	0.305 (0.006)	0.381 (0.024)	0.703 (0.142)	0.398 (0.142)
Information	Arts	0.302 (0.006)	0.399 (0.028)	0.802 (0.094)	0.500 (0.094)
	STEM	0.325 (0.009)	0.317 (0.013)	0.145 (0.072)	-0.180 (0.073)
	Social Sciences	0.374 (0.011)	0.285 (0.020)	0.053 (0.025)	-0.321 (0.027)
Prof. Science & Tech. Services	Arts	0.328 (0.005)	0.338 (0.006)	0.389 (0.047)	0.060 (0.047)
	STEM	0.332 (0.008)	0.333 (0.009)	0.338 (0.069)	0.006 (0.069)
	Social Sciences	0.340 (0.006)	0.328 (0.006)	0.274 (0.039)	-0.066 (0.039)
Transportation & Warehouse	Arts	0.334 (0.004)	0.332 (0.004)	0.298 (0.020)	-0.036 (0.021)
	STEM	0.330 (0.005)	0.336 (0.006)	0.434 (0.032)	0.104 (0.033)
	Social Sciences	0.336 (0.004)	0.332 (0.004)	0.269 (0.031)	-0.067 (0.031)
Wholesale	Arts	0.317 (0.008)	0.344 (0.011)	0.276 (0.103)	-0.040 (0.104)
	STEM	0.379 (0.014)	0.273 (0.021)	0.041 (0.017)	-0.338 (0.021)
	Social Sciences	0.304 (0.008)	0.383 (0.021)	0.683 (0.113)	0.378 (0.113)

Notes: Entries report predicted field-of-study choice probabilities evaluated at low, average, and high levels of local multinational exposure. Values are shown as estimate (standard error). The final column reports within-sample changes from low to high exposure. Standard errors for the difference are computed conservatively as $\sqrt{SE^2_{High} + SE^2_{Low}}$

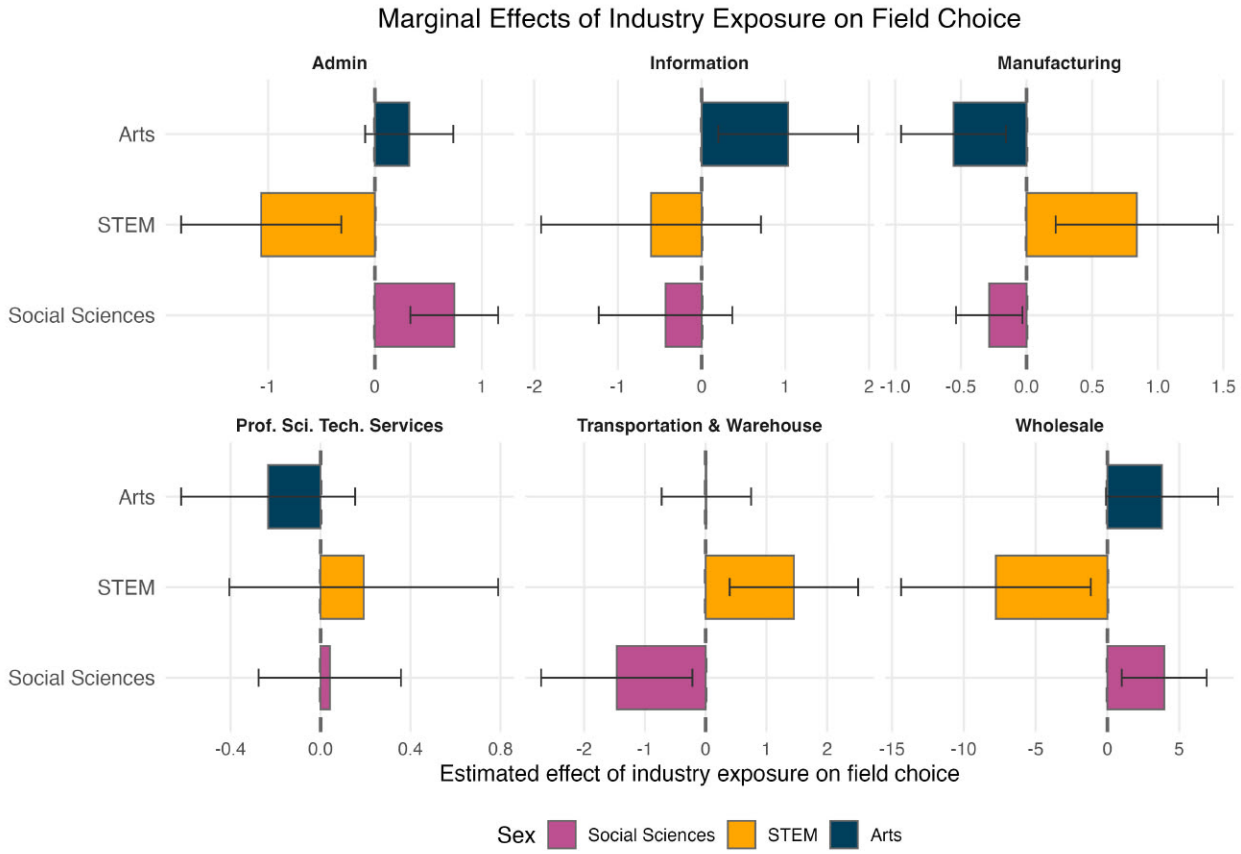


Figure A6: Estimate Results of Main Specification

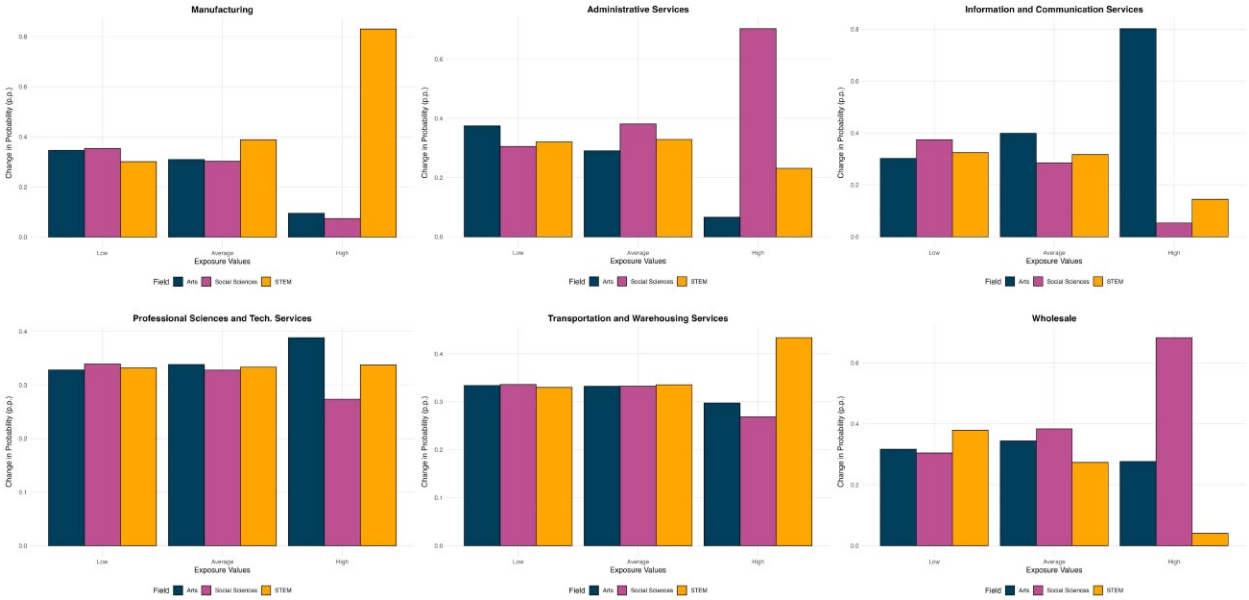


Figure A7: Low to High Exposure Effects for All Industries

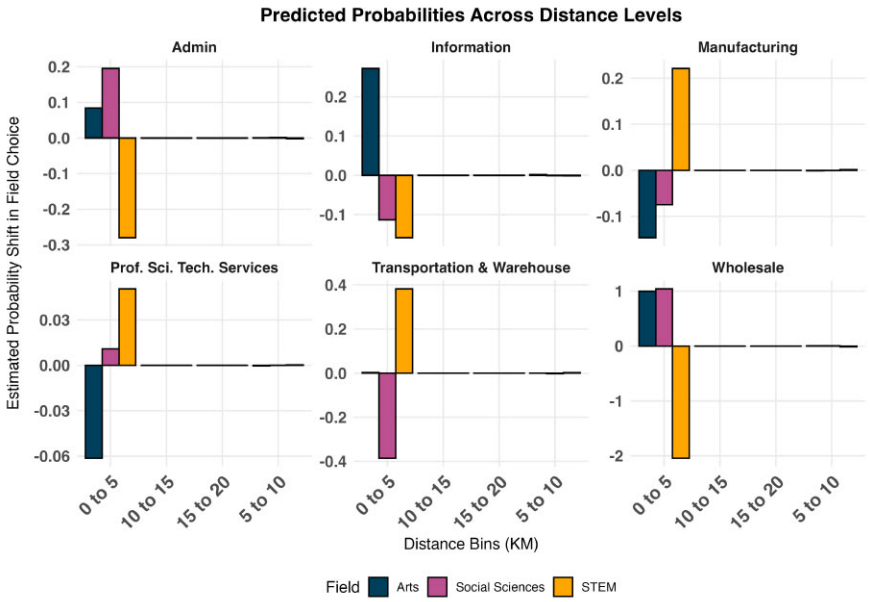


Figure A8: Distance Decay of MNC Effects on Field of Study Choice (Bins)

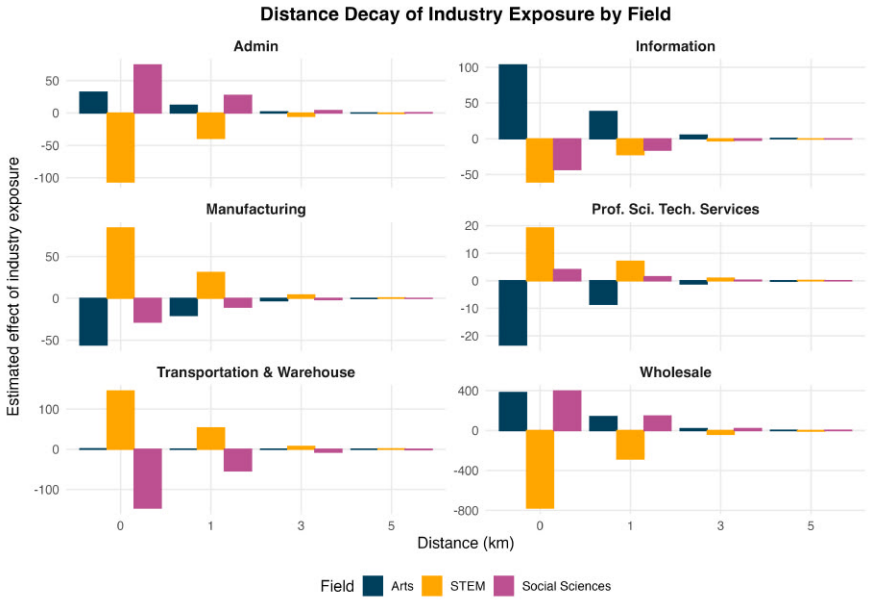


Figure A9: Distance Decay of MNC Effects Within 0 to 5 Km Distances

Table A8: Unweighted MNC Industry Presence Effects on Field of Study Choice

	Field of Study Choice		
	Arts, Writing & Culture	STEM & Applied Sciences	Social Science & Prof. Studies
Administration	-.00014	-0.0002	0.0003**
	(0.0001)	(0.0001)	(0.0001)
Information	0.00012	-0.0005	0.0003
	(0.0002)	(0.0004)	(0.0003)
Manufacturing	-0.0002	0.0007**	-0.0005**
	(0.2038)	(0.0003)	(0.0002)
Prof. Science & Tech. Services	-0.0005**	-0.0014	0.0020***
	(0.0004)	(0.0005)	(0.0005)
Transportation & Warehouse	-0.0014	-0.0004	0.0017
	(0.3755)	(0.0012)	(0.6348)
Wholesale	0.0015	-0.0007	0.0008
	(0.0010)	(0.0014)	(1.5092)
Controls	✓	✓	✓
Canton FE	✓	✓	✓
Year FE	✓	✓	✓
Applications	20,985	92,273	116,904
Pseudo-R2	0.0347	0.0347	0.0347

Statistical significance is displayed as * $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$
Field of Study Fields grouped up as shown in table A2.
Standard errors are clustered at the District level
By definition of AMEs, each row (industry) coefficients should sum to zero

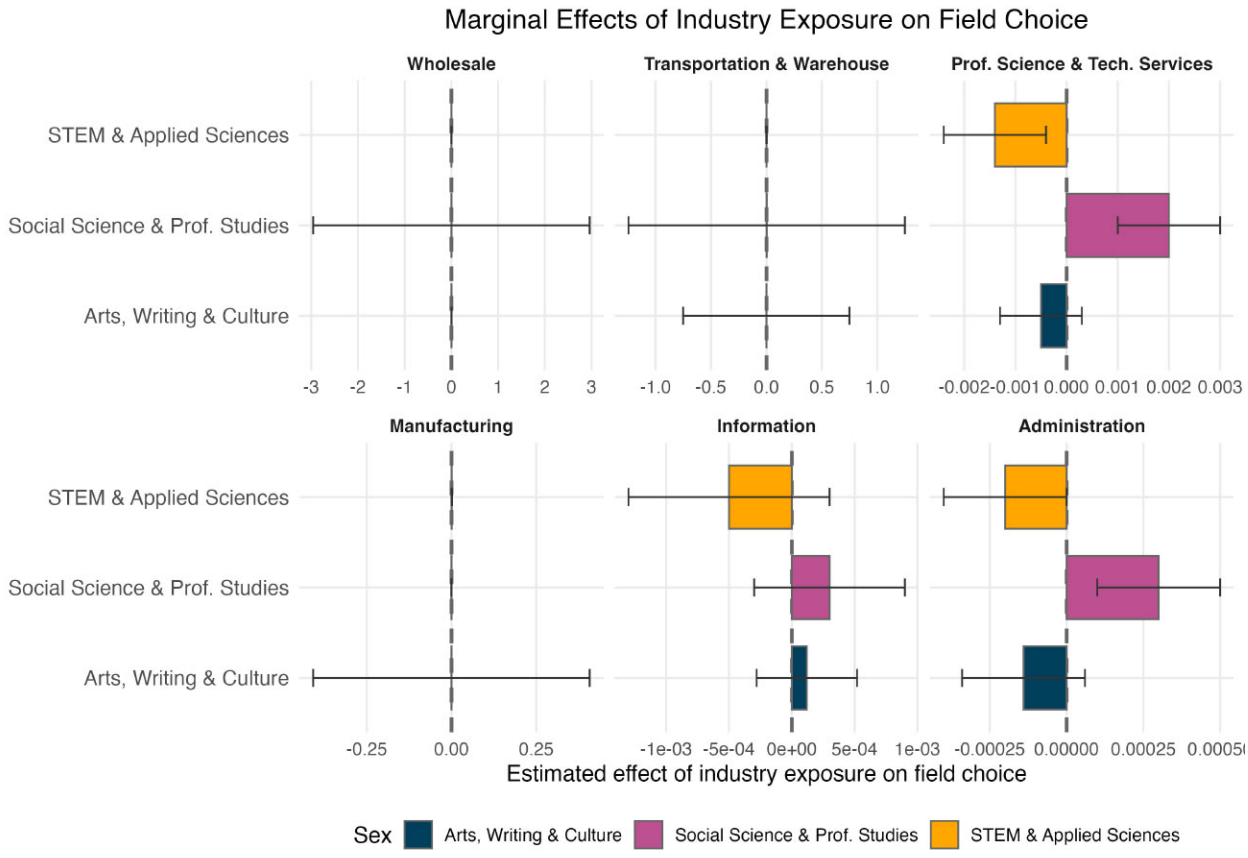


Figure A10: Results of Unweighted Specification

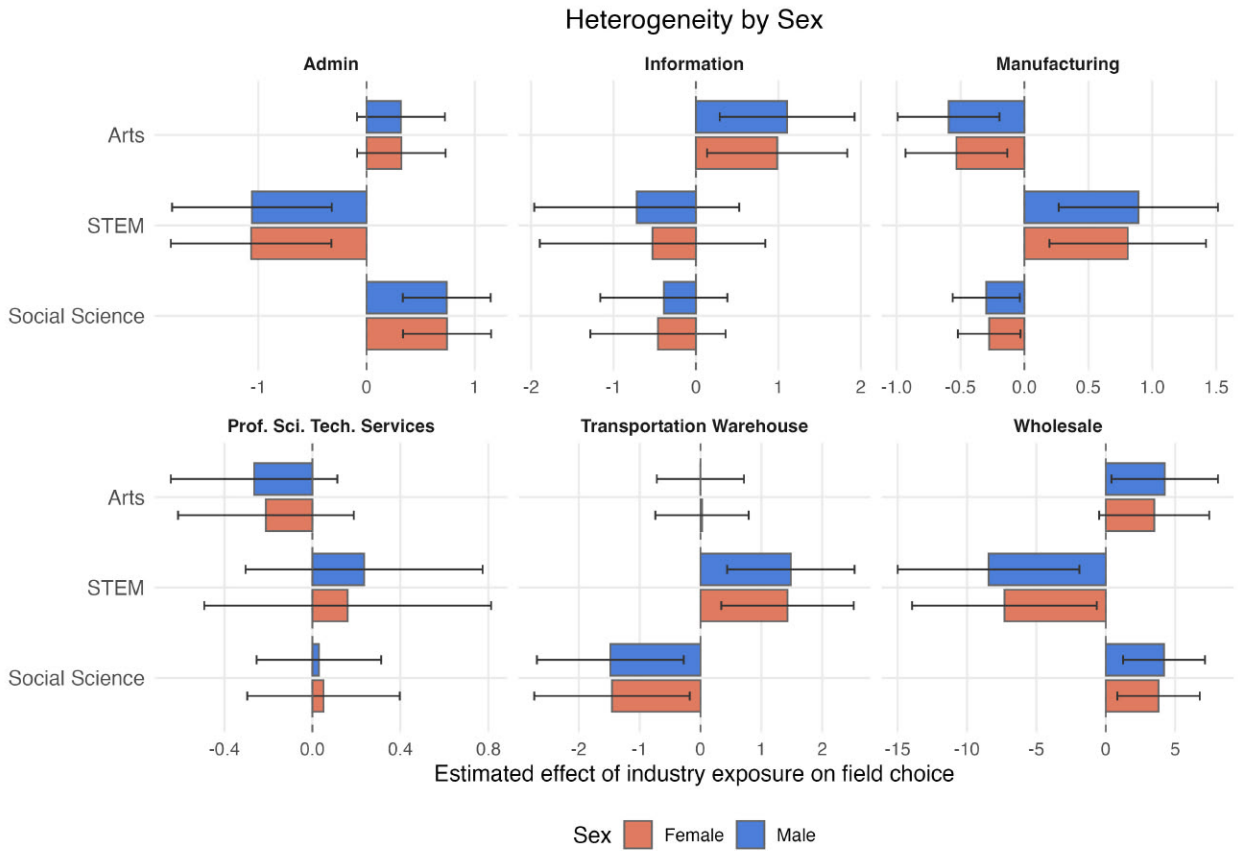


Figure A11: Results of Heterogeneity by Sex

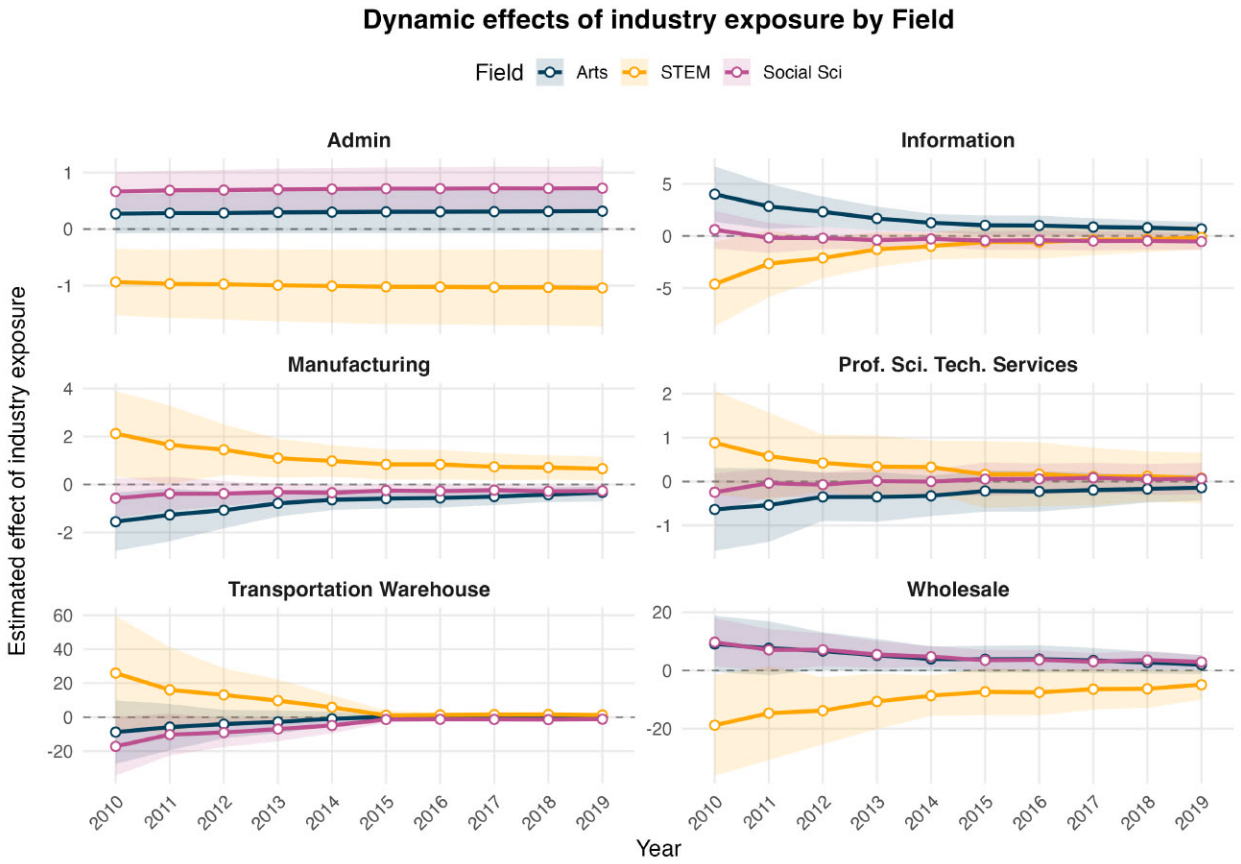


Figure A12: Results of Heterogeneity by Cohort

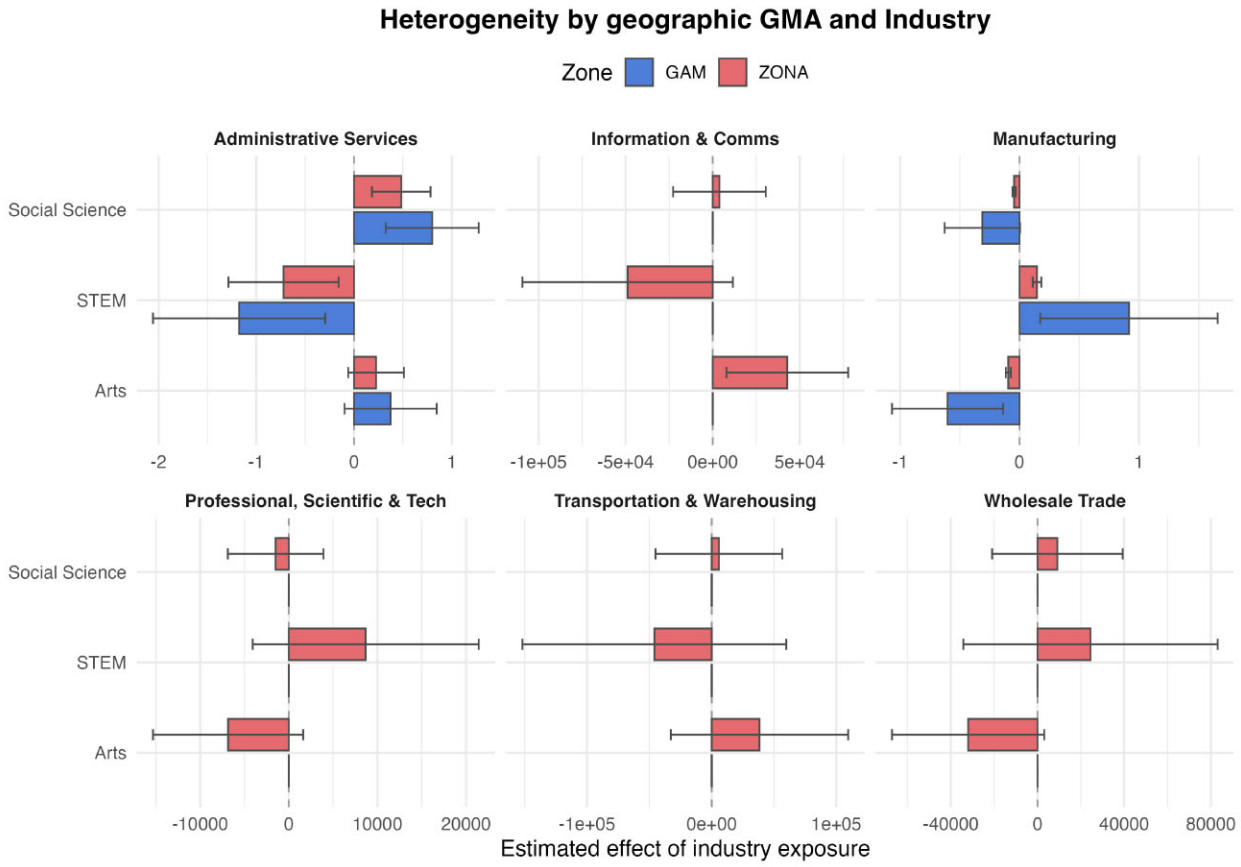


Figure A13: Results of Heterogeneity by Residence in GMA